

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Midterm
Duration: 75 minutes

Semester: Summer 2023
Full Marks: 30

CSE 440: Natural Language Processing II

Figures in the right margin indicate marks.

Name:	ID:	Section:
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Answer all 3

1. [CO1] A. **Explain** with examples: [4]
 - a. Precision
 - b. Micro-average F1
 - c. Macro-average recall
 - d. Statistical significance

2. [CO1] B. You have two classifiers ($y = \sigma(\mathbf{w}^T \mathbf{x} + b)$), classifier A with weight $\mathbf{w}_A = [1 \ 0 \ 2]^T$ and bias $b_A = 1$; and classifier B with weight $\mathbf{w}_B = [1 \ 2 \ 0]^T$ with bias $b_B = 0$. For one example $\mathbf{X} = [1 \ 1 \ 0]^T$, predict y_A and y_B . Which classifier incurs lower cross entropy loss if $\mathbf{X}$'s original label is 1? **Calculate. Show your work.** [6]

2. [CO1] A. Let's say we are working with Shakespeare's plays, and we have [6]
three plays in our hand: Anthony and Cleopatra, Julius Caesar, and Hamlet, with four key characters: Anthony, Brutus, Caesar and Calpurnia. These characters appear in the plays as many times given in table 1. Consider this as your bag-of-words. Now **build** a term-term co-occurrence matrix using this binary bag-of-words. **Explain** what each value in the term-term co-occurrence matrix means.

2. [CO1] B. We have a document that has the word 'shallow' in it 25 times and the [4]
document is 100 words long. If we have a total of 10,000 documents and 1,000 of them have the word 'shallow' in it, calculate the TF-IDF of 'shallow' in that document.

3. [CO2] A. Gender bias in word embeddings means some words are closer to [4]
male words and other words are closer to female words. For example, a poorly trained word vector may associate 'doctor' to 'man' and 'nurse' to 'woman'. If I tell you your embedding has this issue, how will you **test** it? How can you **solve** it? **Explain.**

- B. Explain the training and testing process of the Naive Bayes algorithm [6] for sentiment classification. That is, you are given N sentences where some are positive sentiment sentences and some are negative, and you have to train a Naive Bayes model that will be able to classify a sentence. Then, you got a new sentence which you do not know whether it is positive and negative, and you need to figure out. How are you going to do it?

Table 1

	Anthony	Brutus	Caesar	Calpurnia
Anthony and Cleopetra	16	13	7	0
Julius Caesar	12	9	21	4
Hamlet	0	1	1	0